

Routing Protocols

RIP | OSPF | Path Vector Routing | BGP

Supplementary Notes — Computer Networks CSDC0206

★ These protocols are part of the routing algorithms topic. Exams ask: explain the protocol, compare IGP vs EGP, and discuss advantages/limitations.

Routing protocols can be classified based on where they operate:

Category	Full Name	Operates Within/Between	Examples
IGP	Interior Gateway Protocol	Within a single Autonomous System (AS)	RIP, OSPF, IS-IS, EIGRP
EGP	Exterior Gateway Protocol	Between different Autonomous Systems	BGP (the only widely used EGP today)

Autonomous System (AS): A collection of IP networks and routers under the control of a single organization that presents a common routing policy to the internet. Each AS has a unique AS Number (ASN). Example: Google's AS, AT&T's AS.

1. RIP — Routing Information Protocol

★ RIP is a Distance Vector protocol. Exam questions: explain RIP, its limitations, difference between RIPv1 and RIPv2.

RIP: Routing Information Protocol is one of the oldest and simplest Interior Gateway Protocols (IGP). It is based on the Distance Vector algorithm and uses hop count as its routing metric. Defined in RFC 1058 (RIPv1) and RFC 2453 (RIPv2).

1.1 Key Characteristics of RIP

Feature	Details
Protocol Type	Interior Gateway Protocol (IGP) — Distance Vector
Metric	Hop Count (number of routers a packet must pass through)
Maximum Hop Count	15 hops (16 = infinity = unreachable). This limits RIP to small networks.
Update Timer	Routers send full routing table to neighbors every 30 seconds
Invalid Timer	180 seconds — route marked invalid if no update received
Flush Timer	240 seconds — route removed from table if still no update
Transport Protocol	UDP, Port 520

Algorithm	Bellman-Ford (Distributed)
Convergence	Slow — can take minutes for large networks
Supports VLSM?	RIPv1: No (classful). RIPv2: Yes (classless, sends subnet mask)

1.2 How RIP Works

1. Each router initializes its routing table with directly connected networks (hop count = 0 for self, 1 for direct neighbors).
2. Every 30 seconds, each router broadcasts (RIPv1) or multicasts (RIPv2) its entire routing table to all neighbors.
3. On receiving a neighbor's table, the router applies the Bellman-Ford update rule: if (neighbor's cost to destination + 1) < my current cost, update the route.
4. The process continues until the network converges.
5. If a router does not receive an update for a route within 180 seconds, it marks the route as invalid (metric = 16).
6. After 240 seconds without update, the route is flushed from the routing table.

1.3 RIP Metric — Hop Count

RIP uses hop count as its only metric. Each router (hop) between source and destination adds 1 to the metric.

i Limitation: RIP picks the path with fewest hops, NOT the fastest path. A 3-hop path over gigabit links would lose to a 2-hop path over slow 56Kbps links.

- Maximum useful hop count: 15. A destination with hop count = 16 is considered unreachable.
- This limits RIP to networks with diameter of 15 hops maximum — unsuitable for large enterprise or internet-scale networks.

1.4 RIP Timers

Timer	Default Value	Purpose
Update Timer	30 seconds	Interval between periodic routing table broadcasts
Invalid Timer	180 seconds	If no update received, route is marked invalid (metric=16)
Hold-Down Timer	180 seconds	After a route becomes invalid, ignore better routes briefly to prevent instability
Flush Timer	240 seconds	Route is completely removed from the table after this period without updates

1.5 Problems with RIP

- **Count-to-Infinity:** Same as general DVR problem. Loops increment the metric infinitely.
- **Slow Convergence:** With 30-second update intervals, large network changes take a long time to propagate.
- **Bandwidth Waste:** Broadcasts entire routing table every 30 seconds, even if nothing changed.

- **15 Hop Limit:** Cannot be used in large networks.
- **No Support for Multiple Metrics:** Only uses hop count — ignores bandwidth, delay, reliability.

1.6 Solutions Used in RIP

- Split Horizon: Do not advertise a route back toward the interface it was learned from.
- Poison Reverse: Advertise route back with metric = 16 (infinity) to explicitly signal it is unreachable.
- Triggered Updates: Send updates immediately when a route changes instead of waiting for the 30-second timer.
- Hold-Down Timers: After marking a route invalid, ignore potentially incorrect updates for a period.

1.7 RIPv1 vs. RIPv2

Feature	RIPv1	RIPv2
Standard	RFC 1058	RFC 2453
Addressing	Classful (no subnet mask in updates)	Classless (sends subnet mask — supports VLSM/CIDR)
Update Method	Broadcast (255.255.255.255)	Multicast (224.0.0.9)
Authentication	Not supported	Supported (MD5 password)
Next Hop Field	Not included	Included in updates
Route Tags	Not supported	Supported (for EGP integration)

2. OSPF — Open Shortest Path First

★ OSPF was directly asked in your previous paper. Master the definition, link state concept, areas, metric, and types of links.

OSPF: Open Shortest Path First is a link-state, hierarchical Interior Gateway Protocol (IGP) based on Dijkstra's Shortest Path First (SPF) algorithm. It is widely used in large enterprise networks. Defined in RFC 2328 (OSPFv2 for IPv4) and RFC 5340 (OSPFv3 for IPv6).

2.1 Key Characteristics of OSPF

Feature	Details
Protocol Type	Interior Gateway Protocol (IGP) — Link State
Algorithm	Dijkstra's Shortest Path First (SPF)
Metric	Cost = Reference Bandwidth / Interface Bandwidth (default ref = 100 Mbps)
Convergence	Fast — only sends changes (LSAs), not full tables
Transport Protocol	Directly over IP (Protocol number 89 — not TCP or UDP)
Multicast Addresses	224.0.0.5 (all OSPF routers), 224.0.0.6 (all DR/BDR routers)
Supports VLSM/CIDR?	Yes — classless protocol
Authentication	Yes — plain text or MD5
Scalability	Highly scalable — uses areas to limit LSA flooding
Loop Prevention	Inherent — Dijkstra's algorithm produces a loop-free tree

2.2 Link State Routing — How OSPF Builds Topology

Unlike Distance Vector (which shares routing tables), OSPF routers share Link State Advertisements (LSAs) — packets describing each router's direct connections. Every router eventually has an identical Link State Database (LSDB) of the entire network.

7. Neighbor Discovery: OSPF routers send Hello packets every 10 seconds to discover and maintain neighbor relationships.
8. Database Exchange: New neighbors exchange their LSDBs to synchronize.
9. LSA Flooding: Each router floods its own LSA (describing its links and costs) to every other router in the area.
10. SPF Calculation: Each router independently runs Dijkstra's algorithm on its LSDB to build a shortest path tree rooted at itself.
11. Routing Table Population: Shortest paths are installed as routes in the routing table.
12. Incremental Updates: When topology changes, only the changed LSA is flooded — not the entire database.

2.3 OSPF Areas

OSPF uses a two-level hierarchical structure to improve scalability in large networks. Instead of flooding LSAs across the entire network, they are confined within areas.

Area: A logical grouping of OSPF routers and links. All routers within an area share the same LSDB. LSA flooding is contained within each area, reducing overhead.

- **Backbone Area (Area 0):** The central area to which all other areas must connect directly or via a virtual link. All inter-area traffic passes through Area 0.
- **Regular Areas (Area 1, 2, ...):** Non-backbone areas that connect to Area 0 through Area Border Routers.
- **Stub Area:** Does not receive external routes (routes from outside the AS). Uses a default route instead. Reduces routing table size.
- **Totally Stubby Area:** Cisco extension — receives neither inter-area nor external routes; only a default route.
- **NSSA (Not-So-Stubby Area):** Like a stub area but can import limited external routes using Type-7 LSAs.

OSPF Router Types:

Router Type	Description
Internal Router (IR)	All interfaces in one OSPF area. Maintains one LSDB.
Area Border Router (ABR)	Connects two or more areas. Maintains separate LSDBs for each area. Summarizes routes between areas.
Backbone Router (BR)	Has at least one interface in Area 0.
Autonomous System Boundary Router (ASBR)	Connects OSPF domain to external routing domains (e.g., BGP). Redistributes external routes into OSPF.

2.4 OSPF Metric — Cost

OSPF uses cost as its metric. Lower cost = preferred path.

```
Cost = Reference Bandwidth / Interface Bandwidth  
Default Reference Bandwidth = 100 Mbps (100,000,000 bps)
```

Interface Type	Bandwidth	OSPF Cost
Serial (T1)	1.544 Mbps	64
Ethernet	10 Mbps	10
Fast Ethernet	100 Mbps	1
Gigabit Ethernet	1 Gbps	1 (same as FastEthernet unless ref BW changed)
10 Gigabit Ethernet	10 Gbps	1 (ref BW must be increased)

i Note: For Gigabit and above interfaces, the reference bandwidth should be adjusted (e.g., to 1000 Mbps) so costs are differentiated correctly.

2.5 Types of Links in OSPF

OSPF defines different network types (link types) that affect how Hello packets are sent and whether a Designated Router (DR) is elected.

Link Type	Description	DR/BDR Election?	Example
Point-to-Point	Direct connection between exactly two routers	No	Serial WAN links, PPP
Broadcast Multi-Access	Multiple routers on a shared broadcast medium; DR and BDR elected	Yes	Ethernet LAN
Non-Broadcast Multi-Access (NBMA)	Multiple routers on shared medium without broadcast capability; DR/BDR elected	Yes	Frame Relay, ATM
Point-to-Multipoint	One-to-many connection treated as collection of point-to-point links	No	Hub-and-spoke Frame Relay
Virtual Link	Logical link to connect an area to Area 0 through another area	No	Disconnected areas

2.6 Designated Router (DR) and Backup DR (BDR)

On broadcast networks (like Ethernet), OSPF elects a Designated Router (DR) and Backup Designated Router (BDR) to reduce the number of OSPF neighbor relationships and LSA flooding.

- Without DR/BDR: n routers would form $n*(n-1)/2$ adjacencies — too many on large segments.
- With DR/BDR: All routers form adjacency only with DR and BDR. DR distributes LSAs to all. Reduces overhead significantly.
- DR Election: Based on highest OSPF Priority (configurable, 0-255). Tie-break: highest Router ID (usually highest IP).
- BDR takes over if DR fails — ensuring continuity.

2.7 OSPF vs. RIP

Feature	RIP	OSPF
Type	Distance Vector	Link State
Algorithm	Bellman-Ford	Dijkstra's SPF
Metric	Hop Count (max 15)	Cost (based on bandwidth)
Convergence	Slow (minutes)	Fast (seconds)
Scalability	Small networks only	Large enterprise networks
Hierarchical Design	No	Yes (Areas)
Loop Prevention	Split Horizon, Poison Reverse	Inherent (SPF tree)
Bandwidth Usage	High (full table every 30s)	Low (only changes flooded)
VLSM/CIDR	RIPv1: No, RIPv2: Yes	Yes
Authentication	RIPv2: Yes	Yes

3. Path Vector Routing

★ **Path Vector Routing is the foundation of BGP. Understand how it differs from Distance Vector — it stores the entire path, not just the distance.**

Path Vector Routing: A routing algorithm where each routing table entry contains not just the distance/cost to the destination but the entire path (sequence of routers/ASes) from source to destination. This complete path information enables loop detection and policy-based routing.

3.1 Why Path Vector?

Distance Vector routing suffers from the count-to-infinity problem because routers only know the distance to a destination, not the path. If they knew the full path, they could detect loops immediately (if their own address appears in the path, it's a loop).

Link State routing gives full topology but requires every router to know the complete network map — impractical between different organizations (autonomous systems) due to scale and confidentiality concerns.

Path Vector combines the simplicity of Distance Vector with loop prevention and policy control, making it ideal for inter-AS (internet-scale) routing.

3.2 How Path Vector Routing Works

13. Each router/AS maintains a routing table where each entry contains: Destination, Next Hop, and the complete AS Path (list of ASes to traverse).
14. When a router advertises a route to a neighbor, it appends its own AS number to the AS Path.
15. The receiving router checks the AS Path: if its own AS number appears in the path, it discards the route (loop detected).
16. If no loop, the router may accept and re-advertise the route (with its own AS prepended).
17. Policy rules can be applied to prefer or reject specific paths based on the AS Path content, origin, or other attributes.

3.3 Key Properties of Path Vector

- **Loop Prevention: Built-in** — a router will never accept a route that already includes its own identifier in the path.
- **Policy-Based Routing:** Organizations can choose paths based on business relationships, not just technical metrics.
- **No Count-to-Infinity:** Since routers see the full path, they can immediately reject loops rather than incrementing a counter.
- **Scalability:** Each router only needs to know about its direct neighbors' paths, not the full topology.
- **Path Attributes:** Routes carry rich attributes (AS Path length, origin, local preference, MED, communities) enabling fine-grained routing decisions.

3.4 Path Vector vs. Distance Vector vs. Link State

Property	Distance Vector	Link State	Path Vector
Info Shared	Distance to destination	Full topology (LSAs)	Complete path to destination
Shared With	Direct neighbors only	All routers (flooded)	Direct neighbors only

Loop Detection	Indirect (split horizon, timers)	Inherent (SPF tree)	Direct (check AS Path)
Convergence	Slow	Fast	Moderate
Scalability	Small networks	Medium networks (with areas)	Internet-scale (BGP)
Policy Support	None	Limited	Rich (BGP attributes)
Example	RIP	OSPF, IS-IS	BGP

4. BGP — Border Gateway Protocol

★ **BGP is THE routing protocol of the internet. It uses Path Vector. Exams ask: what is BGP, how does it work, what are its attributes, and iBGP vs eBGP.**

BGP: Border Gateway Protocol is the standardized Exterior Gateway Protocol (EGP) used to exchange routing and reachability information between Autonomous Systems (ASes) on the internet. It is the routing protocol that makes the internet work. Current version: BGP-4 (RFC 4271).

i BGP is often called the 'protocol of the internet.' It connects all the ISPs, cloud providers, and large organizations that form the internet backbone.

4.1 Key Characteristics of BGP

Feature	Details
Protocol Type	Exterior Gateway Protocol (EGP) — Path Vector
Purpose	Route exchange between Autonomous Systems (inter-domain routing)
Transport	TCP, Port 179 (reliable transport for large routing tables)
Metric / Decision	Path Attributes (AS Path, Local Preference, MED, etc.) — not a single metric
Convergence	Slow by design — stability and policy correctness prioritized over speed
Scalability	Internet-scale — handles ~1 million+ routes (full internet routing table)
Route Updates	Incremental — only sends changes after initial full table exchange
Authentication	MD5 hash over TCP session
Session Type	iBGP (within AS) and eBGP (between ASes)

4.2 BGP Session Types

- **eBGP (External BGP):** BGP session between routers in **DIFFERENT** autonomous systems. This is the standard internet BGP. Peers are typically directly connected.
- **iBGP (Internal BGP):** BGP session between routers **WITHIN** the same autonomous system. Used to propagate BGP routes throughout an AS. Requires all iBGP routers to be fully meshed (or use Route Reflectors).

Feature	eBGP	iBGP
Peers	Different ASes	Same AS
TTL	1 (peers must be directly connected by default)	255 (can be multiple hops apart)
AS Path modification	Appends own AS number	Does not modify AS Path
Next Hop	Router's own IP	Unchanged (next hop of eBGP peer preserved)
Loop Prevention	AS Path contains own AS = loop	Split horizon rule (don't re-advertise to iBGP)

4.3 BGP Operations — How BGP Works

Step 1: TCP Session Establishment

BGP peers establish a TCP connection on port 179. Unlike OSPF (which uses IP directly) or RIP (UDP), BGP uses TCP for reliability — routing tables are too large to risk packet loss.

Step 2: BGP Message Exchange

Message Type	Purpose
OPEN	First message after TCP connection. Exchanges BGP version, AS number, Router ID, and Hold Time.
KEEPALIVE	Sent every 1/3 of Hold Time (default every 60s) to maintain the session. If not received within Hold Time (default 180s), session is dropped.
UPDATE	Carries routing information: new reachable prefixes (NLRI) with attributes, and withdrawn routes (unreachable prefixes).
NOTIFICATION	Sent when an error is detected. Causes the BGP session to be closed.

Step 3: BGP Finite State Machine

Each BGP session goes through states: Idle → Connect → Active → OpenSent → OpenConfirm → Established.

- Idle: BGP is waiting to start. No TCP connection.
- Connect: BGP is trying to establish TCP connection.
- Active: TCP connection failed; BGP is retrying.
- OpenSent: TCP connected; OPEN message sent; waiting for peer's OPEN.
- OpenConfirm: OPEN received and valid; exchanging KEEPALIVE.
- Established: Full BGP session up. UPDATE messages can now be exchanged.

4.4 BGP Path Attributes

BGP does not use a single metric like hop count or cost. Instead, routes carry Path Attributes that BGP uses to select the best path. These attributes also enable policy-based routing.

Attribute	Type	Description
AS_PATH	Well-Known Mandatory	List of AS numbers the route has traversed. Used for loop detection and path selection (shorter = preferred).
NEXT_HOP	Well-Known Mandatory	IP address of the next-hop router to reach the destination.
ORIGIN	Well-Known Mandatory	How the route was originated: IGP (i), EGP (e), or Incomplete (?). IGP preferred.
LOCAL_PREF	Well-Known Discretionary	Used within an AS (iBGP) to indicate preferred exit point. Higher = preferred. Default = 100.
MED (MULTI_EXIT_DISC)	Optional Non-Transitive	Hints to neighboring AS about preferred entry point. Lower = preferred.

ATOMIC_AGGREGATE	Well-Known Discretionary	Indicates route has been aggregated (summarized).
COMMUNITY	Optional Transitive	Group of routes sharing a common property. Used for tagging and policy application.
WEIGHT	Cisco Proprietary	Local to a single router (not advertised). Higher = preferred.

4.5 BGP Best Path Selection Algorithm

When BGP has multiple paths to the same destination, it uses the following decision process (in order):

18. Highest WEIGHT (Cisco-specific, local only)
19. Highest LOCAL_PREF (preferred exit from AS)
20. Route originated locally (via network or redistribute command)
21. Shortest AS_PATH length
22. Lowest ORIGIN code (IGP < EGP < Incomplete)
23. Lowest MED value
24. eBGP path preferred over iBGP path
25. Lowest IGP metric to the NEXT_HOP
26. Lowest Router ID (tie-breaker)

★ **Memory trick for BGP selection: 'We Love Oranges As Oranges Mean Pure Refreshment' — Weight, Local Pref, Originate, AS Path, Origin, MED, Prefer eBGP, Router ID.**

4.6 BGP and the Internet

- The global internet routing table (carried by BGP) contains over 900,000+ prefixes as of 2024.
- Every ISP, cloud provider (AWS, Google, Azure), CDN, and large organization has one or more BGP ASes.
- BGP is what allows you to reach any website anywhere in the world — it stitches together thousands of independently managed networks.
- BGP hijacking is a security concern where a malicious AS advertises someone else's IP prefixes, redirecting traffic.
- RPKI (Resource Public Key Infrastructure) is a security framework designed to prevent BGP hijacking.

4.7 Limitations of BGP

- Slow Convergence: BGP prioritizes stability — route changes can take minutes to propagate across the internet.
- Complexity: BGP configuration and troubleshooting requires deep expertise.
- Security: BGP has no built-in authentication for route origins (addressed partially by RPKI).
- iBGP Full Mesh: All iBGP routers must peer with each other — scales poorly in large ASes (solved by Route Reflectors or Confederations).
- Memory and CPU: Full internet routing table requires significant router resources.

5. Complete Comparison — All Four Protocols

Feature	RIP	OSPF	Path Vector	BGP
Category	IGP	IGP	Concept (used by BGP)	EGP
Algorithm Type	Distance Vector	Link State	Path Vector	Path Vector
Scope	Within AS	Within AS	Between ASes	Between ASes
Metric	Hop Count (max 15)	Cost (bandwidth-based)	Path attributes	Multiple attributes (AS Path, Local Pref, etc.)
Transport	UDP Port 520	IP Protocol 89	—	TCP Port 179
Convergence	Slow	Fast	Moderate	Very Slow (by design)
Loop Prevention	Split Horizon, Poison Reverse	Dijkstra SPF tree	AS Path check	AS Path check
Scalability	Very small (15 hops max)	Large (with areas)	Internet-scale	Internet-scale (millions of routes)
Policy Support	None	Limited	Full	Full (rich attributes)
Authentication	RIPv2: MD5	MD5	—	MD5
Hierarchy	No	Yes (Areas)	—	Yes (ASes)
Update Method	Full table every 30s	Incremental LSAs	Incremental	Incremental after initial exchange
RFC	RFC 2453	RFC 2328	—	RFC 4271

6. Quick Definitions for Exam

Term	Exam-Ready Definition
RIP	A Distance Vector IGP that uses hop count (max 15) as metric. Routers share full tables every 30 seconds via UDP port 520. Suffers from slow convergence and count-to-infinity.
OSPF	A Link State IGP using Dijkstra's SPF algorithm. Routers flood LSAs to build a complete topology map and calculate shortest paths. Uses cost (bandwidth-based) as metric. Hierarchical (areas). Fast convergence.
Path Vector Routing	A routing approach where each route advertisement includes the complete path (sequence of ASes). Enables loop detection (reject route if own AS in path) and policy-based routing. Basis of BGP.
BGP	The internet's inter-domain (EGP) routing protocol. Uses TCP port 179. Exchanges routes between Autonomous Systems. Uses path attributes (AS Path, Local Pref, MED) for route selection. Path Vector algorithm.

AS (Autonomous System)	A collection of IP networks and routers under one organization with a common routing policy. Identified by a unique ASN. BGP routes between ASes; IGP (RIP, OSPF) route within an AS.
LSA	Link State Advertisement — packet flooded by OSPF routers describing their links and costs. All routers in an area maintain an identical LSDB built from LSAs.
eBGP	External BGP — BGP session between routers in DIFFERENT autonomous systems. Standard internet BGP.
iBGP	Internal BGP — BGP session between routers within the SAME autonomous system. Distributes BGP routes internally.
AS_PATH	A BGP attribute listing all AS numbers a route has traversed. Used for loop detection and path length comparison in best-path selection.
DR (Designated Router)	Elected OSPF router on broadcast networks that distributes LSAs to all others, reducing the number of adjacencies needed.
Hold-Down Timer (RIP)	After a route becomes invalid, the router ignores new information about that route for 180 seconds to prevent instability.
LSDB	Link State Database — maintained by each OSPF router; contains all LSAs received; represents the complete network topology within an area.

Good luck — you know this now!